Field	Value
Document No.	WI-003
Title	Manual Mill Setup and Operation
Revision	B
Effective Date	2026-08-10
Prepared By	Owner
Approved By	Jacob Honick, Owner
Revision Approval Date	2026-11-05
Next Review	2027-08-10

WI-003 Manual Mill Setup and Operation

1. Scope

Applies to MAN-MILL-001, the Grizzly G0704 manual mill in the single Torchworx LLC shop. The owner confirms the actual machine, available tooling, workholding and measurement methods for each job. Optional accessories and automated features are not assumed.

2. Preparation and Safety

1. Review the current drawing, purchase order and QF-004 traveler. Confirm that the features can be produced and verified with owner-confirmed available equipment.
2. Check maintenance and machine condition under QP-019. Resolve conditions affecting safe operation or conformity before use.
3. Clean mounting surfaces and select suitable workholding. Verify secure grip, support, alignment and tool clearance for the actual tolerances.
4. Wear suitable eye protection, secure loose clothing and hair, and keep hands clear of rotating tools. Stop the spindle before setup changes, tool changes, chip removal or measurement. Remove drawbar wrenches and setup tools before starting. Do not leave the running machine unattended.

3. Operation

1. Verify head/workholding alignment using an owner-selected method appropriate to the job. Record significant setup criteria in the job notes.
2. Select suitable tools, speeds, feeds and cut sequence using manufacturer information and general industry practice as guidance, within the machine's actual capability.
3. Install tools manually with the spindle stopped. Establish datums using an available owner-confirmed method; verify position and clearance before cutting.
4. Produce and fully inspect the first piece under WI-005 before continuing. Verify the next piece after a change that could affect conformity.
5. Perform full inspection by default under QP-006 and QP-007, recording results on QF-005/QF-006. A different inspection extent requires a customer-specified or customer-approved written plan.
6. Stop for any abnormal condition or nonconformance, hold affected product under QP-008, and correct/re-verify before continuing.
7. Deburr and clean parts as required, sign off QF-004, apply status tags under WI-008 and clean the machine safely. Record maintenance needs.

4. References

- QP-005 Production Planning and Control; QP-006 In-Process Inspection; QP-007 Final Inspection and Product Release
- QP-008 Nonconforming Product Control; QP-019 Preventive Maintenance
- QF-004 Job Traveler; QF-005 In-Process Inspection Record; QF-006 Final Inspection Report; QF-027 Equipment List
- WI-005 First Piece Inspection; WI-008 Part Marking

5. Revision History

Rev	Date	Description	Approved By
A	2026-08-06	Initial release	Shop Owner

Revision B Approval

Approved 2026-11-05 by Jacob Honick, Owner. Reflect the Grizzly G0704 manual mill and owner-confirmed methods without assumed accessories.